1. Quadratic Formula

$$ax^2 + bx + c = 0 \quad \Rightarrow \quad x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

2. Finite Series of Payments

$$PV = \sum_{t=1}^{N} \frac{C}{(1+r)^t} = \frac{C}{r} - \frac{1}{(1+r)^N} \cdot \frac{C}{r} = \left(1 - \frac{1}{(1+r)^N}\right) \cdot \frac{C}{r}$$

3. Finite Series of Payments with Growth

$$PV = \frac{C}{1+r} + \frac{C \cdot (1+g)}{(1+r)^2} + \frac{C \cdot (1+g)^2}{(1+r)^3} + \ldots + \frac{C \cdot (1+g)^{N-1}}{(1+r)^N} = \left(1 - \left(\frac{1+g}{1+r}\right)^N\right) \cdot \frac{C}{r-g}$$

4. Infinite Series of Payments

$$PV = \frac{C}{1+r} + \frac{C \cdot (1+g)}{(1+r)^2} + \frac{C \cdot (1+g)^2}{(1+r)^3} + \ldots = \sum_{t=1}^{\infty} \frac{C \cdot (1+g)^{t-1}}{(1+r)^t} = \frac{C}{r-g}$$

5. Continuous Returns

$$r^* = ln(1+r) \quad \Leftrightarrow \quad 1 + r = e^{r^*} = \exp(r^*)$$

6. Expected Value

$$\hat{\mu} = \frac{1}{N} \cdot \sum_{t=1}^{N} x_t$$

7. Variance and Covariance

Sample

$$\hat{\sigma}^2 = \frac{1}{N-1} \cdot \sum_{t=1}^{N} (x_t - \mu_x)^2$$

$$Cov[x_t, y_t] = \frac{1}{N-1} \cdot \sum_{t=1}^{N} (x_t - \mu_x)(y_t - \mu_y)$$

Population

$$\hat{\sigma}^2 = \frac{1}{N} \cdot \sum_{t=1}^{N} \left(x_t - E(\tilde{x})\right)^2$$

$$Cov[x_t, y_t] = \frac{1}{N} \cdot \sum_{t=1}^{N} \left(x_t - E(\tilde{x})\right)\left(y_t - E(\tilde{y})\right)$$

8. Portfolio Variance

$$Var_{Portfolio} = \sum_{i=1}^{N} \sum_{j=1}^{N} x_i x_j \sigma_{ij}$$